

Rectangles

Statement

You have N **red** rectangles. Rectangle i has height h_i and width w_i . The N rectangles have their bottom side on the x-axis and are placed side by side (red rectangle i and $i + 1$ share a side).

You are required to draw at most K blue rectangles such that every **red** rectangle is inside of **exactly one blue** rectangle. You are able to determine the positions and the dimensions of the **blue** rectangles freely. Hence, you want to find the minimum total area of blue rectangles to cover all N **red** rectangles.

Input format

The first line consists two integers N and K ($2 \leq N \leq 200$) ($1 \leq K \leq 10^9$).

N lines follow, the i -th line contains two integers h_i and w_i ($1 \leq h_i, w_i \leq 1000$).

The bounds and points of each task are given below:

Task 1 (15 points): $N \leq 7, K \leq 7$

Task 2 (20 points): $N \leq 200, K = 1$

Task 3 (20 points): $N \leq 200, K = 2$

Task 4 (15 points): $N \leq 200, K = 998244353$

Task 5 (30 points): $N \leq 200, K \leq N - 1$

Output format

One integer denoting the minimum area of blue rectangles to cover all N **red** rectangles using at most K **blue** rectangles as described in the statement.

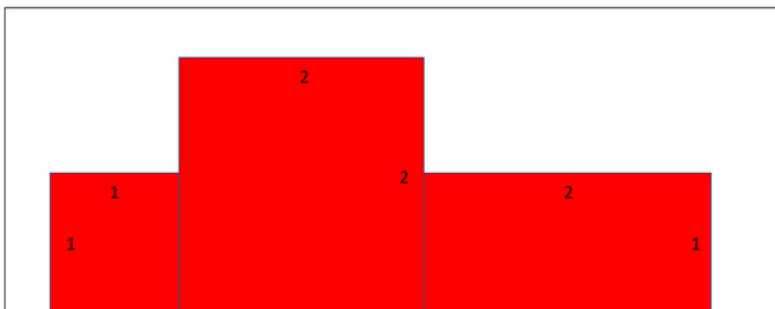
Sample Input

```
3 2
1 1
2 2
1 2
```

Sample Output

8

Explanation: The diagram looks like this:



Note that you cannot make the following arrangement of blue rectangles:



This is an invalid arrangement, as **red** rectangle 2 is covered by both **blue** rectangle 1 and **blue** rectangle 2. A **red** rectangle can only be inside one **blue** rectangle only.

The answer can be obtained by the following arrangement:

